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Adoption of artificial intelligence in academic libraries: A systematic review of current practices, challenges, and research opportunities

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ABSTRACT

Given the strategic role of academic libraries in higher education, this study provides a systematic literature review on the current practices, challenges to the adoption of Artificial Intelligence and the future research directions in academic libraries. A total of 29 relevant articles were used for review based on their relevance. Only peer-reviewed journal articles and conference papers published in the last five years (2020–2024) were selected and included in the study. Web of Science, SCOPUS, and Library and Information Science Abstract (LISA) were the databases consulted. The current practices were grouped into six major dimensions, including technical services, references and information services, administrative services and circulation, collection development, information literacy and user education, and professional development and collaboration. The results showed that the most prominent use of AI in academic libraries is for reference and information services. Furthermore, it was found that 48.28 % (14) of the articles cited Funding as the major institutional challenge to AI adoption in academic libraries. A cluster of studies included in the review emphasized the need for more user-driven research on AI applications in academic libraries. This study reflects the importance of retraining librarians to develop new competencies in managing AI-enhanced communication tools, including chatbot integration and prompt engineering.

Introduction

Academic libraries have consistently been at the forefront of adopting emerging technologies to enhance their operations and improve user experience (Boateng, 2024). Over the years, these libraries have integrated various digital tools to support research, learning, and information access. Among the technologies reshaping academic library functions today is Artificial Intelligence (AI). AI represents a broad field of computer science focused on developing systems capable of performing tasks that typically require human intelligence, such as learning, reasoning, problem-solving, and decision-making (Alkatheiri, 2022). The integration of AI into library environments has enabled librarians to automate repetitive processes, enhance data-driven decision-making, and provide more personalized and efficient information services (Ali et al., 2021; Echedom & Okuonghae, 2021; Yusuf et al., 2022). Within the operations and services of libraries in recent times, AI has become instrumental in transforming traditional operations such as cataloguing, classification, information retrieval, and user engagement. It supports reference services, enhances digital resource discovery, improves administrative efficiency, and contributes to professional development among librarians (Cox, 2024; Zondi et al., 2024). Through these applications, AI technologies are redefining how academic libraries meet the information needs of diverse users and how library professionals develop new competencies in an increasingly data-driven environment. Conceptually, AI in this study refers to the broad framework of intelligent systems that simulate aspects of human cognition to improve efficiency and innovation within academic libraries. This understanding indicates a wide spectrum of AI-driven applications, from automation and predictive analytics to intelligent information systems. While the field of AI continues to evolve rapidly, the focus of this review remains on the general adoption of AI technologies in academic libraries as discussed within recent scholarly literature. Although a number of systematic reviews have examined AI integration in libraries (Concha et al., 2024; Harisanty et al., 2025; Mwantimwa & Msoffe, 2025), but a few have explored in detail the current practices, categories of challenges, and future research needs from a global perspective. From

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available literature, Mabona et al. (2024) examined AI adoption in academic libraries without emphasizing specific thematic challenges, while Echedom and Okuonghae (2021) focused mainly on African contexts. Similarly, Zondi et al. (2024) reviewed AI in academic library services but did not address future research directions. Given these gaps, this study becomes essential due to the strategic role of academic libraries in higher education and the growing need to address the limitations of existing literature. The aim of this paper is to provide a systematic review of current practices, challenges, and future research directions relating to the adoption of Artificial Intelligence in academic libraries. A systematic review of this nature not only serves as a gauge of available literature on current practices in academic library contexts, but it also gives impetus for future research. It provides evidence-based data on the prevailing challenges of AI adoption on a global scale, which will guide academic librarians and researchers to prepare for future challenges.

Research questions

The systematic literature eview is guided by the following questions:

1. What are the current practices of AI in Academic libraries?
2. What types of academic library challenges (technology, organizational, environment, and individual) of AI have been reported in the literature?
3. What future research directions related to AI in academic libraries have been reported in the literature?

Methodology

The researchers conduct a literature review using the Preferred Reporting Items for the Systematic Review and Meta-analysis (PRISMA) guidelines. Moher et al. (2009) explained that PRISMA helps in systematically reporting evidence-based items, which assists in a rigorous appraisal of the literature. They argued that PRISMA has four aspects in a study, namely identification, screening, eligibility, and additional studies. Systematic literature reviews have been increasingly applied in information and library science disciplines, as evident in studies by Lee et al. (2023) and Mabona et al. (2024).

Databases and selection criteria

This research utilized English-language articles and conference papers published in reputable journals. This also informed the choice of databases. The researchers conducted an initial pilot search of the databases to determine whether the articles were relevant. While this systematic literature review focuses on Generative Artificial Intelligence (Gen AI), studies addressing broader Artificial Intelligence (AI) were also included where relevant. This was necessary because Gen AI, which gained prominence after the public release of ChatGPT in late 2022, evolved from earlier AI frameworks. Many papers published between 2020 and 2022 discussed AI applications that directly underpin current generative technologies. Therefore, the essence of including these studies indicates a comprehensive and holistic understanding of how academic libraries have transitioned from general AI adoption to the present era of Gen AI integration. The focused databases were three: (i) Web of Science Core Collection (Science Citation Index Expanded, Social Science Citation Index, Social Science Citation Index (SSCI) Conference Proceedings Citation Index - Science, Conference Proceedings Citation Index - Science, Conference Proceedings Citation Index - Social Science and Humanities, Emerging Sources Citation Index). (ii) Scopus database and (iii) the Library and Information Science Abstract (LISA). While these databases are well-established in LIS as information sources, they were specifically chosen because they contain peer-reviewed articles and meet international standards. These databases were searched from April 2nd, 2025.

Search strategy, inclusion and exclusion of articles

The article search was limited to peer-reviewed articles published between 2020 and 2024, as they were deemed most relevant to the study. Data was downloaded from the three databases and collated on an online Google Sheet. Search terms for Scopus, LISA databases include TITLE-ABS-KEY (“generative AI” OR “generative artificial intelligence” OR “artificial intelligence” OR “LLM” OR “Large Language Model” OR “ChatGPT” OR “Gemini”) AND (Librar* OR “Information Servic*”) and search term for Web of Science database TS (“generative AI” OR “generative artificial intelligence” OR “artificial intelligence” OR “LLM” OR “large language model” OR “ChatGPT” OR “Gemini”) AND (Librar* OR “Information Servic*”). While the whole the search process was rigorous and iterative, the researchers limited the filter to Library and Information Science publications to retrieve specific articles from Library and Information Science abstracts related to AI and Gen AI. This strategy facilitated the retrieval of relevant articles.

The initial search was 319 journal articles and conference papers (peer-reviewed), which were extracted into a Google Excel sheet. The PRISMA information flow diagram for the search protocol was adopted (Faulkner & Reiter, 2025). After removing duplications and triplications, the researchers collectively reviewed the titles and abstracted the articles by screening 221. In the next stage, the researchers systematically reviewed the complex text of 80 articles and excluded one article that was not retrieved. Additionally, 51 articles did not cover publications related to academic libraries, and AI challenges, practices, and future trends were excluded. The researchers relied on Liu and Liu (2023) checklist for quality assessment, which includes five areas: research objectives, research methods, discussion of findings, research questions, and suggestions for future research. The researchers assessed these against the quality assessment and gave 2 points per area, while some received 1 point based on the evaluation (Yes (2 points), Partial (1 point), and No (0 points)). The researchers included articles that scored five or higher (on a 10-point scale) in the study. The authors resolved conflicts vis-à-vis inclusion and exclusion criteria based on relevance, year of publication, authenticity of the reported data (peer-reviewed and conference proceedings), and language used. Researchers ensure the validity and accuracy of search data by cross-checking. The three researchers performed these processes together and repeated to ensure data accuracy and reliability. The researchers conducted the screening and eligibility assessments three times within a week, and they met a third time to complete the process. The final papers considered for final analysis and synthesis were 29 articles (see Fig. 1).

Inclusion and exclusion criteria

1. The articles included were based on AI and academic libraries as research objects. In contrast, articles not related to AI and academic libraries were excluded such as AI in public libraries and AI in school libraries contexts.
2. Articles included captured challenges, practices, and future trends, while those that are not related to these areas were excluded.
3. The researchers searched only for academic, peer-reviewed articles as the quality standard, and included conference proceedings.
4. Inclusion of articles were in line with the quality assessment criteria.

Results

Research characteristics

As shown in Table 1, the study comprises 29 articles published between 2020 and 2024. Holistically, this study consists of 28 peer-reviewed journals and one conference paper. None of the papers used adopt a mixed-method approach. Further analysis of the papers (see Fig. 2) indicates that more than half of the articles 15 (51.72 %) were published in 2024; this reveals an increase in trends of relevant

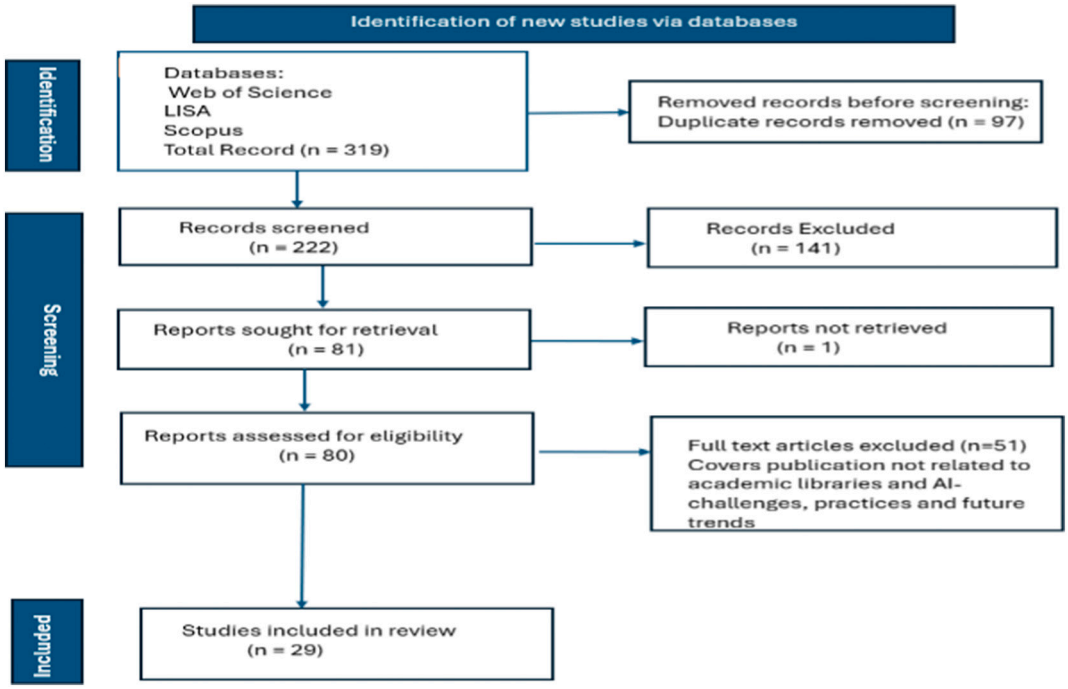


Fig. 1. Procedure for Inclusion and Exclusion.

publications and exponential growth of research in this subject area. The Journal of Academic Librarianship, Library Philosophy and Practice have the highest number of articles, with both having 3, respectively, constituting 10.3 % of the total articles. College and Research Libraries, Information Technology and Libraries, Library Hi Tech News, New Review of Academic Librarianship, and South African Journal of Libraries and Information Science have 6.9 % of the articles used (see Fig.3).

Results on current practices of AI in academic libraries

This section (Table 2) analyzed the current practices of AI in academic libraries, based on data extracted from 29 selected studies. These practices were grouped into six major dimensions, including technical services, references and information services, administrative services and circulation, collection development, information literacy and user education, and professional development and collaboration. The prevalence of each practice was quantified, and specific examples were drawn from the literature. One of the most prominent dimensions observed was technical services, where AI has been applied to enhance back-end library operations. Among these, cataloguing services were the most frequently mentioned practice, appearing in 27.58 % (8) of the reviewed studies (Abba, 2024; Ali et al., 2021; Enakrire & Oladokun, 2024; Isiaka, 2023; Li & Coates, 2024; Lund et al., 2020; Winkler & Kiszl, 2022). Similarly, classification tasks (24.13 %) were also reported in studies where AI tools helped streamline subject categorization processes (Cox, 2024; Enakrire & Oladokun, 2024). Indexing (10.34 %) and intelligent content extraction and curation (3.45 %) were relatively less reported (Isiaka, 2023; Enakrire & Oladokun, 2024; Hosseini & Holmes, 2023; Okunlaya et al., 2022). Other niche practices, such as automation, AI meta generation, acquisition assistance, and rare resource curation, each appeared in a single study (3.45 %), indicating emerging experimentation in these areas. These findings suggest that AI has begun to transform traditional technical operations in libraries.

The dimension of reference services also demonstrated notable integration of AI, particularly in answering directional and ready reference questions (Ali et al., 2021; Çakmak & Eroğlu, 2024; Enakrire & Oladokun, 2024; Hosseini & Holmes, 2023; Isiaka, 2023; Lund et al., 2020; Winkler & Kiszl, 2022). In addition to this, a significant body of literature emphasized the role of reference and information services through AI in providing chat transcription and expedited reference services, referenced in 27.58 % of the studies (Çakmak & Eroğlu, 2024; Li & Coates, 2024; Okunlaya et al., 2022; Shahzad et al., 2024; Wang, 2022). Following this, studies indicated that Gen AI appears to be relevant to users in terms of web search optimization, and information/library search (Çakmak & Eroğlu, 2024; Wang, 2022).

Literature indicates that AI facilitates discovery systems and intelligent database search for users (Ali et al., 2021; Okunlaya et al., 2022). Findings further report that AI enhances search, engagement, and streamlines research (Meakin, 2024). Other practices such as information services, responding to students' queries, answering directional or ready reference questions, recommending materials to patrons, personalizing learning resources, retrieving academic materials faster, predicting future resource needs, helping users to locate books, and selective dissemination of information (SDI) were reported in studies (Abba, 2024; Enakrire & Oladokun, 2024; Lund et al., 2020; Shahzad et al., 2024; Rabatseta et al., 2024; Winkler & Kiszl, 2022). These findings suggest that AI is helping libraries scale personalized reference interactions.

The scope of administrative services and circulation revealed several innovative uses of AI aimed at boosting organizational efficiency. Handling complex tasks (6.89 %) and speeding up decision-making processes (6.89 %) were emphasized in studies such as Yusuf et al. (2022) and Jan et al. (2024), where AI was utilized to inform resource planning and policy updates. Findings also highlighted AI as a tool for marketing library resources, circulation, managing workload efficiency, improving library service productivity, replacing time-consuming tasks, easing routine administrative burdens for university librarians, protecting user data privacy, analyzing library statistics, and recommending acquisitions based on usage patterns (Abba, 2024; Isiaka, 2023; Lund et al., 2020; Shahzad et al., 2024; Zondi et al., 2024). Security-related applications, including Radio Frequency Identification (RFID) for check-in/check-out, self-check-out machines for books, resource security, and facial recognition for access control, were featured in 24.13 % of the studies, particularly in Ali et al. (2021) and Çakmak and Eroğlu (2024).

In the area of collection development, literature found relatively

Table 1
Details of selected publication included in the study.

No	Year	Authors	Title	Journal	Method
1	2024	Zondi, Nombuso Phamela; Epizitone, Yugioh; Nkomo, Ntando; Mthalane, Peggy Pinky; Moyane, Smangele; Luthuli, Mthokozisi; Khumalo, Mbalenhle; Phokoje, Samkelisiwe	A review of artificial intelligence implementation in academic library services.	South African Journal of Libraries & Information Science	Qualitative design
2	2022	Yusuf, T. I., Adebayo, O. A., Bello, L. A., & Kayode, J. O.	Adoption of artificial intelligence for effective library service delivery in academic libraries in Nigeria	Library Philosophy & Practice	Qualitative design
3	2023	Mustapha, Abdrahman A.; Yusuf, Ibrahim O.	Adoption of artificial intelligence to improve library service delivery in academic libraries in Kwara State, Nigeria	Library Philosophy & Practice	Qualitative design
4	2023	Isiaka, Abdullahi Olayinka	Application and Use of Artificial Intelligence (AI) for Library Services Delivery in Academic Libraries in Kwara State, Nigeria.	Library Philosophy & Practice	Quantitative study-Questionnaire
5	2021	Ali, Muhammad Yousuf; Naeem, Salman Bin; Bhatti, Rubina	Artificial Intelligence (AI) in Pakistani university library services.	Library Hi Tech News	Qualitative Study-Case stud
6	2024	Cox, Andrew M.	Artificial Intelligence and the academic library.	Journal of Academic Librarianship	Qualitative design (Review)
7	2024	Enakrire, Rexwhite Tega; Oladokun, Bolaji David	Artificial intelligence as enabler of future library services: how prepared are librarians in African university libraries.	Library Hi Tech News	Qualitative design
8	2023	Huang, Yingshen; Cox, Andrew M.; Cox, John	Artificial Intelligence in academic library strategy in the United Kingdom and the Mainland of China.	Journal of Academic Librarianship	Qualitative design
9	2024	Dalton, Courtney	Artificial Intelligence, Language, and Humanization in the Academic Library.	Theological Librarianship	Qualitative design
10	2023	Yang, Sharon Q.; Mason, Sarah	Beyond the Algorithm: Understanding How ChatGPT Handles Complex Library Queries.	Internet Reference Services Quarterly	Descriptive (Case study)
11	2024	Jan, Saeed Ullah; Khan, Muhammad Sajjad Ali; Khan, Ali Saeed	Organizational Readiness to Adopt Artificial Intelligence in the Library and Information Sector of Pakistan.	Evidence-Based Library & Information Practice	Qualitative study -Interview
12	2020	Lund, Brady D.; Omame, Isaiah; Tijani, Solomon; Agbaji, Daniel	Perceptions toward Artificial Intelligence among Academic Library Employees and Alignment with the Diffusion of Innovations' Adopter Categories.	College & Research Libraries	Quantitative study - Questionnaire
13	2022	Ajani, Yusuf Ayodeji; Tella, Adeyinka; Salawu, Khadijah Yetunde; Abdullahi, Firdausi	Perspectives of Librarians on Awareness and Readiness of Academic Libraries to Integrate Artificial Intelligence for Library Operations and Services in Nigeria.	Internet Reference Services Quarterly	Quantitative study - Questionnaire
14	2023	Hosseini, Mohammad; Holmes, Kristi	The Evolution of Library Workplaces and Workflows via Generative AI.	College & Research Libraries	Qualitative design
15	2021	Echedom, Anthonia U.; Okuonghae, Omorodion	Transforming academic library operations in Africa with artificial intelligence: Opportunities and challenges: A review paper.	New Review of Academic Librarianship	Qualitative design
16	2024	Atwood, Gary S.	Try AI Day: Introducing Library Faculty and Staff to Artificial Intelligence Tools Through Hands-On Experimentation.	Journal of Electronic Resources in Medical Libraries	Descriptive (Case study)
17	2024	Abba, Tukur	Use of Artificial Intelligence Technologies in Rendering Library Services: An Empirical Evidence from University Libraries in Africa.	African Journal of Library, Archives & Information Science	Qualitative study
18	2024	Johnson, Stacy; Owens, Erin; Menendez, Hannah; Kim, Dianna	Using ChatGPT-generated essays in library instruction.	Journal of Academic Librarianship	Descriptive (Case Study)
19	2024	Rabatseta, Phuti Concilia; Modiba, Mashilo; Ngulube, Patrick	Utilization of artificial intelligence for the provision of information services at the University of Limpopo libraries.	South African Journal of Libraries & Information Science	Qualitative study (Content analysis)
20	2022	Winkler, Bea; Kiszl, Péter	Views of Academic Library Directors on Artificial Intelligence: A Representative Survey in Hungary.	New Review of Academic Librarianship	Quantitative Study - questionnaire
21	2024	Meakin, L	Exploring the Impact of Generative Artificial Intelligence on Higher Education Students' Utilization of Library Resources A Critical Examination	Information Technology and Libraries	Qualitative study (Content analysis)
22	2024	Li, LL; Coates, K	Academic library online chat services under the impact of artificial intelligence	Information discovery and delivery	Qualitative study (Content analysis)
23	2022	Okunlaya, RO; Abdullah, NS; Alias, RA	Artificial intelligence (AI) library services innovative conceptual framework for the digital transformation of university education	Library Hi Tech	Quantitative study - Questionnaire
24	2024	Çakmak, T., & Eroğlu, Ş	The use of artificial intelligence in university libraries in Turkiye: Practices, and perspectives of library directors	Information Development	Qualitative design (Review)
25	2023	Cox, A	How artificial intelligence might change academic library work: Applying the competencies literature and the theory of the professions	Journal of the Association for Information Science and Technology	Quantitative study - Questionnaire
26	2024	Shahzad, K; Khan, SA; Iqbal, A; Javeed, AMD	Identifying university librarians' readiness to adopt artificial intelligence (AI) for innovative learning experiences and smart library services: an empirical investigation	Global Knowledge, Memory and Communication	Qualitative design
27	2024	Mabona A.; Van Greunen D.; Kevin K.	Integration of Artificial Intelligence (AI) in Academic Libraries: A Systematic Literature Review	2024 IST-Africa Conference, IST-Africa 2024	Qualitative design

(*continued on next page*)

Table 1 (*continued*)

No	Year	Authors	Title	Journal	Method
28	2022	Wang Y.	Using Machine Learning and Natural Language Processing to Analyze Library Chat Reference Transcripts	Information Technology and Libraries	Qualitative study (Case study)
29	2023	Lappalainen, Yrjo; Narayanan, Nikesh	Aisha: A Custom AI Library Chatbot Using the ChatGPT API.	Journal of Web Librarianship	Qualitative design

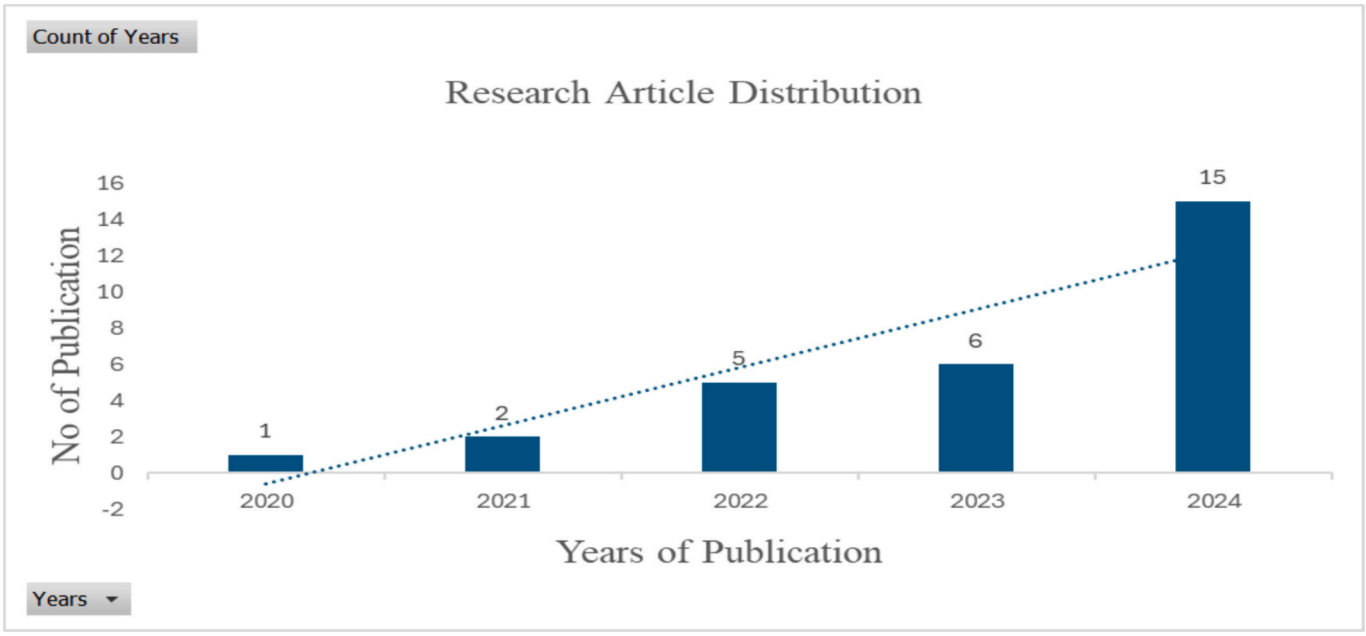


Fig. 2. Distribution of articles by year of publication.

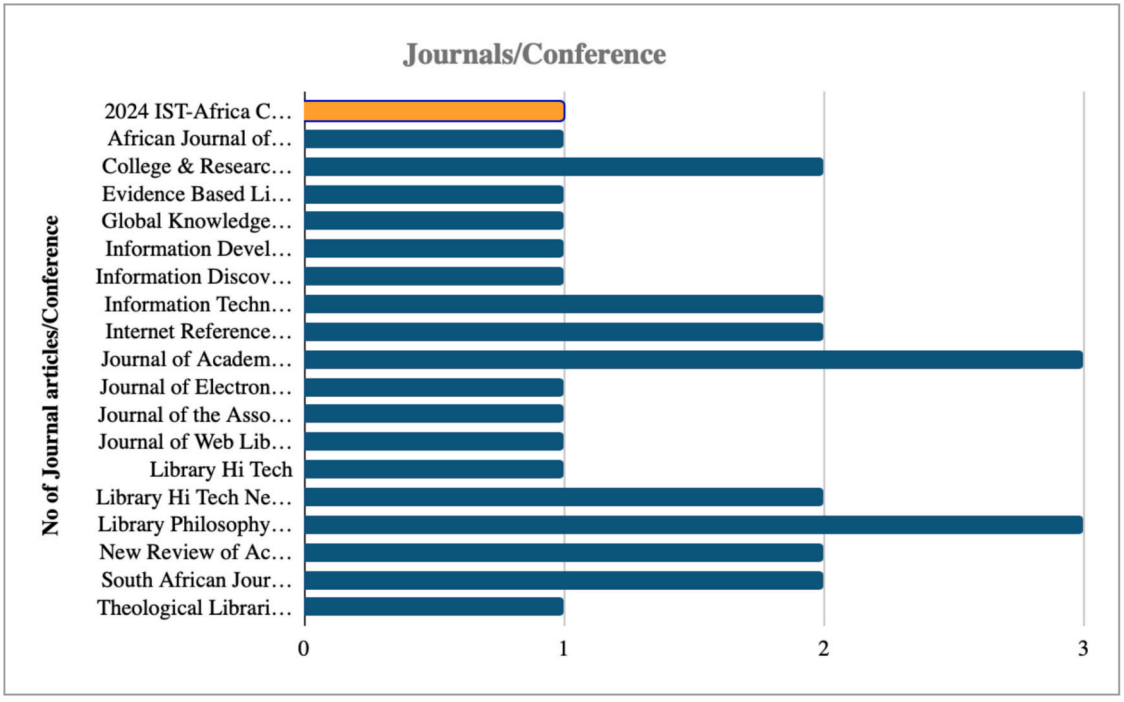


Fig. 3. Number of articles included in major journals/conferences.

Table 2
Current practices of AI in academic libraries.

Dimensions	Practices	Article no	Number of articles	Proportion (%)
Technical services	Automation	14	1	3.45 %
	Indexing	4, 7, 14	3	10.34 %
	Catalogue/ cataloguing service	4, 5, 7, 12, 20, 22, 24, 17	8	27.58 %
	Classification	4, 5, 7	3	24.13 %
	Acquisition	5	1	3.45 %
	Intelligent content extraction and curation	23	1	3.45 %
	Curation of rare resources	26	1	3.45 %
	AI meta generation	6	1	3.45 %
References and information services	Answering directional and ready reference questions	4, 5, 7, 12, 14, 17, 19, 20, 24, 26	9	34.48 %
	Helping users to locate books	19	1	3.45 %
	Discovery systems	5	1	3.45 %
	Web search optimization	22, 24, 25	3	10.34 %
	Information and library search	12, 14, 16, 18, 10, 23	6	20.69 %
	Recommending materials to patrons	12	1	3.45 %
	Information service	19, 20	2	6.89 %
	AI for Selective Dissemination of Information (SDI)	7	1	3.45 %
	Chat transcription and expedited reference services	5,8, 20,22,23, 24, 26, 28	8	27.58 %
	Personalization of learning resources	26	1	3.45 %
	Retrieve academic materials faster	26	1	3.45 %
	Predict future resource needs	26	1	3.45 %
Administrative services and circulation	Self-check-out machine for books	17	1	3.45 %
	Handling complex tasks	2, 3	2	6.89 %
	Circulation services	24, 5	2	6.89 %
	Intelligence acquisition	23	1	3.45 %
	Used as a marketing tool for the library	17	1	3.45 %
	Expedite decision-making processes	11, 12	2	6.89 %
	Replace time-consuming tasks	11	1	3.45 %
	A tool for statistics evaluation and recommendations	17	1	3.45 %
	Enhancing library service efficiency	1, 4	2	6.89 %
	Enhance user-centric service	26	1	3.45 %
	Facial recognition for access	5	1	3.45 %
	Alleviate the burden of routine administrative tasks for University Librarians	26	1	3.45 %
	Security and privacy of users' data	4, 26	2	6.89 %
	RFID for self-check-in/out and security,	5, 24, 8	3	10.34 %
	Manage workload efficiently	26	1	3.45 %
	Help librarians analyze usage patterns and resource preferences,	26	1	3.45 %
	Assisting in the charging and discharging of library materials	17	1	3.45 %
Collection Development	Collection development	20, 7	2	6.89 %
	Quick and instant support for librarians to users' queries	26	1	3.45 %
	Use of intelligent virtual agents	8	1	3.45 %
	User authentication	5	1	3.45 %
Information Literacy and User Education	Teaching and advocating cautious and reflective adoption of AI technologies in libraries	9	1	3.45 %
	Library-hosted AI events	8	1	3.45 %
	Guide users to retrieve relevant resources	22, 26	2	6.89 %
	Promotion, orientation, and training	24	1	3.45 %
Professional Development and Collaboration	Professional development by providing unique trends in research in LIS	26	1	3.45 %
	Establish collaboration by sharing platforms and repositories with librarians	26	1	3.45 %

limited engagement with AI. Collection development support was referenced in just 6.89 % of the studies, with contributions from Winkler and Kiszl (2022) and Enakrire and Oladokun (2024). Other practices, such as the use of intelligent virtual agents, authentication, and providing quick and instant support for librarians to users' queries, were mentioned only once each, suggesting that these are still emerging practices (Ali et al., 2021; Dalton, 2024). Regarding information literacy and user education, several studies have documented proactive efforts by libraries to guide users in retrieving relevant resources using AI technologies (Shahzad et al., 2024). Finding also reported teaching and advocating cautious and reflective adoption of AI technologies in libraries (Dalton, 2024). Library-hosted AI events, library instruction, promotion, orientation, and training were also reported in the literature (Huang et al., 2023; Li & Coates, 2024). Finally, the literature reveals the use of AI technologies for professional development and collaboration (Shahzad et al., 2024). As a result, the findings suggest that AI plays a role in facilitating professional growth by tracking global research trends and promoting collaborative platforms among librarians.

Results on challenges faced by academic libraries with the adoption and use of AI

Academic libraries and librarians face numerous challenges and barriers when adopting AI. Scholars have identified several challenges, which the researchers categorized into technical, organizational, environmental, and individual challenges. This is related to the Technology-Organization-Environment (TOE) framework by Tornatzky and Fleischer (1990). This theory emphasizes understanding the challenges of organizations based on three key factors: technical, organizational, and environmental challenges. Academic libraries are also intellectual organizations.

Technical challenges

Researchers from a systematic review-based study identified the need to address technical challenges, in addition to the challenges of AI in academic libraries and librarianship, to achieve the goals of AI technologies in academic libraries. Out of 29 studies analyzed, as shown in Table 3, 17 studies explained technical challenges, which account for 58.62 %. This highlights two significant technical challenges, including gaps in technological skills and knowledge, and challenges in data quality. The gaps in technological skills and knowledge include limited IT skills and expertise among librarians, a lack of comprehensive knowledge and instruction on AI, and the integration of AI into existing systems are the prevalent technological challenges. The gaps in technological skills and knowledge were also identified (Ajani et al., 2022; Ali et al., 2021; Echedom & Okuonghae, 2021; Huang et al., 2023; Isiaka, 2023; Mabona et al., 2024; Shahzad et al., 2024; Zondi et al., 2024). This shows that academic librarians do not have sufficient skills and knowledge of AI. Additionally, other scholars have highlighted the challenges of data quality, including inaccurate or confusing bot responses, insufficient datasets, and deficiencies in the quality of bibliographic data as technical challenges to AI adoption in academic libraries (Lappalainen & Narayanan, 2023, Yang & Mason, 2023; Çakmak & Eroğlu, 2024, Dalton, 2024).

Institutional challenges

A total of 29 studies were analyzed to explore the institutional-level challenges encountered in the implementation and adoption of AI within academic libraries. The data were thematically grouped into four major categories, including funding, infrastructural issues, ethical concerns, and organizational issues. The analysis reveals that while enthusiasm for AI integration is growing, substantial institutional barriers continue to impede its effective deployment in many academic contexts. Funding emerged as the most frequently cited challenge, with 14 of the 29 studies (48.28 %) reporting funding as a critical barrier to AI adoption in

Table 3
Challenges faced by academic libraries with the adoption and use of AI.

Dimensions	Challenges	Article no	Number of articles	Proportion (%)
Technical challenges				
Technical skill and knowledge gap	Limited IT skills and expertise among librarians	1, 4,5, 8, 13, 15, 26, 27	8	27.5 %
	Lack of complete knowledge and instruction of AI	12, 17, 29	3	10.3 %
	Integration of AI into an existing system	6, 8	2	6.89 %
Data quality	Hallucination - inaccurate or confusing bot responses	10, 29	2	6.0.89 %
	Insufficient datasets and deficiencies in the quality of bibliographic data	9, 24	2	6.89 %
	Institutional challenges			
Finance	Funding	1, 2,3,4, 5,8, 11, 13,15, 17, 21, 24, 26, 27	14	48.28 %
	High Cost of Technology	21	1	3.45 %
Institutional infrastructural issue	Limited power supply	13, 21, 27	3	10.34 %
	Poor networking	21	1	3.45 %
	Outdated technologies	21	1	3.45 %
	Inadequate infrastructure	1, 2, 3, 15, 24, 26	6	20.69 %
	Lack of full-fledged AI tools	5	1	3.45 %
Ethical concerns	AI ethics - academic integrity, biased and false information, intellectual property and plagiarism, Copyright issues, Privacy concerns, Cybersecurity	12, 22, 22, 24, 27, 26, 22	7	24.13 %
	Ethical concerns	6, 8,	2	6.89 %
Organizational issues	Organizational technological practices	11	1	3.45 %
	Lack of institutional policy	26	1	3.45 %
	High authority support	26	1	3.45 %
	leadership/management adaptation	6	1	3.45 %
	Inequality in library development	5	1	3.45 %
	Lack of human-AI integration	7	1	3.45 %
	Environmental Challenges			
External Issues	Robot uprising -	20	1	3.45 %
	Misleading public understanding of AI capabilities	9	1	3.45 %
	Governance disparities globally	6	1	3.45 %
	Exposure to International Standards	15	1	3.45 %
Individual Challenges				
Competences	Lack of confidence in new technology	11	1	3.45 %
	An increase in the margin of error resulting from excessive automation	20	1	3.45 %
	Over-reliance on AI	10	1	3.45 %
	Over-trusting AI outputs due to anthropomorphic language	9	1	3.45 %
	Librarian preparedness	6, 7	2	6.89 %
Individual fears and concerns	Fear of job loss	10, 2, 3, 6	4	13.79 %
	Job insecurity concerns	5	1	3.45 %
	Resistance to change	15, 21	2	6.89 %
	Traumatisation of librarians	20	1	3.45 %
	Redundancy of librarians' role	20	1	3. 45 %
	Negative perception from librarians	20	1	3.45 %
	Risk aversion	8	1	3.45 %

academic libraries. Several studies have noted that many academic institutions, particularly in developing countries, struggle to allocate dedicated financial resources for AI technologies (Çakmak & Eroğlu, 2024; Cox, 2024; Echedom & Okuonghae, 2021; Mabona et al., 2024; Meakin, 2024; Moustapha & Yusuf, 2023; Shahzad et al., 2024; Yusuf et al., 2022; Zondi et al., 2024).

These constraints not only affect the procurement of AI systems but may also hinder training, maintenance, and continuous development. Supporting this, findings indicated high cost of technology (Meakin, 2024) as a specific obstacle, stating that even where there is institutional will, the financial burden often outweighs available resources. Also, closely tied to funding is the issue of institutional infrastructural issues cited in 12 studies (41.38 %). These limitations encompass a range of deficiencies, including inadequate infrastructure (Çakmak & Eroğlu, 2024; Echedom & Okuonghae, 2021; Moustapha & Yusuf, 2023 & Shahzad et al., 2024), limited power supply (Ajani et al., 2022; Mabona et al., 2024; Meakin, 2024; Yusuf et al., 2022; Zondi et al., 2024) as the major problems affecting the full implementation of AI in academic libraries. Similarly, other studies have highlighted issues with outdated technologies, poor networking (Meakin, 2024), and a lack of comprehensive AI tools (Ali et al., 2021).

Ethical challenges were identified in approximately one-third of the reviewed studies (31.03 %), making them the third most prominent institutional concern. The studies revealed multifaceted ethical dilemmas surrounding AI, including issues of academic integrity, misinformation, intellectual property, privacy concerns, bias, plagiarism, copyright issues, and cybersecurity (Çakmak & Eroğlu, 2024; Li & Coates, 2024; Lund et al., 2020; Mabona et al., 2024; Okunlaya et al., 2022; Shahzad et al., 2024). Moreover, several studies have raised further ethical concerns (Cox, 2024; Huang et al., 2023).

Although each was cited by only one study (3.45 %), various organizational issues collectively revealed a fragmented institutional readiness for AI integration. These included the lack of institutional policies and high authority support (Shahzad et al., 2024), low levels of leadership adaptation (Cox, 2024), and insufficient organizational technological practices (Jan et al., 2024). Other studies noted inequalities in library development across institutions (Ali et al., 2021), limited human-AI integration (Enakrire & Oladokun, 2024), and the influence of university size on readiness (Jan et al., 2024).

Environmental and individual challenges

A few studies, comprising 13.8 % of the reviewed articles, indicated that environmental challenges also act as bottlenecks to the implementation of AI in academic libraries. The four articles identified in this category (Cox, 2024; Dalton, 2024; Echedom & Okuonghae, 2021; Winkler & Kiszl, 2022) reported robot uprising, misleading public understanding of AI capabilities, global governance disparities on how AI should be implemented and exposure to international standards as external issues hindering full adoption of AI in academic libraries settings. Although each of these challenges was respectively, 3.45 % of the studies, their influence can have a significant negative impact on how AI technologies are adopted in academic libraries and by academic librarians for delivering information services to patrons, if not given sufficient attention in the literature.

It was equally clear that, in addition to other challenges to AI implementation in academic contexts, individual challenges have been specifically highlighted in the literature. As Table 3 indicates, the analysis of studies on individual challenges reveals competence and individual fears as the significant challenges faced by academic libraries. The articles revealed that fear of job loss (Cox, 2024; Meakin, 2024; Moustapha & Yusuf, 2023; Yusuf et al., 2022) and resistance to change (Echedom & Okuonghae, 2021; Meakin, 2024) are the most cited individual challenges identified in this category. Other factors identified included the traumatization of librarians, redundancy in librarians' roles, a negative perception among librarians (Winkler & Kiszl, 2022), and risk aversion (Huang et al., 2023). Overall, 11 out of the 29 articles explicitly reviewed address individual fears and concerns as prominent challenges that must be addressed in the shortest possible time for the full implementation of AI technologies in academic libraries. Thus, this further reveals a critical knowledge gap in the literature, especially as the call for libraries to adopt AI technologies continues to gain momentum in library and information science literature.

Future studies and research direction as identified in the literature

To ensure AI adoption in libraries and information services, most studies have identified specific areas for future directions. Jan et al. (2024) advocate for future studies on AI competence and existing infrastructure in libraries, and studies that examine the readiness of libraries toward AI adoption across various geographical contexts or within smaller universities in Pakistan. The authors also make a case for a future comparative study to explore the positive perception of AI and the experiences of implementing AI in libraries. Similarly, as a way of enabling libraries to take full advantage of the opportunities and the affordances of AI technologies, future studies are to focus on AI applications in library services, attitude of librarians in the use of AI applications, and sustainability of AI applications in academic libraries (Çakmak & Eroğlu, 2024).

Furthermore, a cluster of studies emphasized the need for more user-driven research on AI applications in libraries. For instance, Wang (2022) studied chat transcripts in an academic library with the aim of building a model to enhance reference services. The scholar proposed a future research direction that focuses on developing multi-category models and using machine learning to support information seeking, citation, and answering of reference questions, and printing of relevant resources in libraries. In a more recent study, Johnson et al. (2024) and Dalton (2024) suggested, respectively, that there is a need for future studies to address the concerns of critical and literacy skills on the use of AI, as well as research on library instruction involving AI. While these research directions largely have important implications for library users, they further justify the need for future research on full-scale adoption of AI in academic libraries (Ali et al., 2021). Academic libraries are adopting GenAI tools to enhance digital literacy and support research and information processes (Yeung et al., 2025).

Furthermore, for AI applications in academic libraries to achieve their desired purpose, Winkler and Kiszl (2022) established that there is a need for studies that consider a deep analysis of AI applications issues in libraries. Huang et al. (2023) propose a robust methodological approach in future research that combines interviews and surveys to examine the perspectives of library directors on AI applications in academic libraries. As a way to ensure that libraries, organizations, and librarians fully understand how AI is developing, the scholars further make recommendations for surveys on current practices on AI adoption in academic libraries in the United Kingdom and China, and how libraries develop their strategies in the context of their institution and national policy trends. On ethics and the responsible use of AI, Lappalainen & Narayanan (2023) argue that there are still ethical and legal issues regarding the adoption of AI in academic libraries, making it imperative for academic librarians to explore new ways to provide information services while also being mindful of the ethical implications.

Discussion of findings

The study is a systematic literature review on the adoption of AI in academic libraries, focusing on current practices, challenges, and future directions. Discussing the three research questions of the study and the findings, the researchers contribute to the overall literature on the adoption of AI in academic libraries:

RQ1. *What are the current practices of AI in Academic libraries?*

From the findings, it can be deduced that various services and

practices in academic libraries have been enhanced or have the potential to be enhanced with AI. With the thematic identification of technical services, reference and information services as well as administrative and circulation services, as critical areas where AI technologies have impacted academic libraries, one can argue that academic libraries around the world have much to gain from the adoption and full implementation of artificial intelligence. However, the onus is on librarians and other information professionals to ensure that the classification of library resources and answering reference questions, having been revealed as a major area of application, are efficiently done by leveraging AI technologies. Academic librarians would also need to rely greatly on AI technologies for collection development, which may include, but are not limited to, selection and acquisition of library resources and evaluation of resources to effectively cater to users' needs and queries. It behoves librarians, researchers, and students to take full advantage of the possibilities of AI use in academic libraries, especially as literature indicates that adoption of artificial intelligence also enhances information literacy, user education, professional development, and collaboration in academic libraries.

RQ2. *What types of academic library challenges (technology, organizational, environment, and individual) of AI have been reported in the literature?*

Adoption of AI in academic libraries is growing. However, scholars are skeptical about the adoption due to the various challenges surrounding AI. It was found that academic libraries face four primary challenges when adopting AI: technical, institutional, environmental, and individual challenges. To address these challenges, most studies have established that institutional challenges are the primary concern, with finance as the most significant concern. Other institutional challenges include ethical concerns, institutional infrastructural issues, and organizational issues. Additionally, technical challenges, the limited IT skills and expertise among librarians are the most critical. This is similar to the Gmiterek and Kotuła (2025) study, which established that academic libraries lack staff competencies in AI. In addressing environmental challenges and individual challenges, the primary concern is the fear of job loss.

To address these challenges, several studies have proposed solutions to technical issues, including the training and retraining of librarians on the use of AI for service delivery, as well as the provision of suitable AI infrastructure by the government and relevant agencies (Moustapha & Yusuf, 2023; Shahzad et al., 2024). Also, librarians should have up-to-date knowledge on the developments of Gen AI (Bates, 2023). Funding is a central institutional challenge in the study; to address the challenge, scholars have proposed solutions such as sustainable funding models that will enhance AI's expansion, upgrading, and maintenance. Academic libraries should seek external funding, including grants (Shahzad et al., 2024).

RQ3. *What are the future research directions related to AI in academic libraries as reported in the literature?*

Future studies can help to understand the field, prepare for future research, identify gaps in the literature, and provide future direction for the field of study. While there are a lot of studies on AI adoption in academic libraries, the researchers identified four areas for future research directions: the readiness for AI adoption (Çakmak & Eroğlu, 2024), skills in AI usage (Dalton, 2024; Johnson et al., 2024), research, ethical and legal issues (Lappalainen & Narayanan, 2023), and the integration of multi-category models to provide holistic services in academic libraries (Wang, 2022). The study suggests that future studies should investigate academic librarians' readiness to adopt and implement AI. This calls for seminars, workshops and practical training, retraining and continuous training for academic librarians on adoption, adaptation and implementation of AI. This will help in the implementation of holistic AI tools to enhance academic library practices. Future studies have proposed multi-category models that can be applied to various practices in academic libraries. Additionally, ethical and legal concerns are crucial for future studies. Research on legal and ethical issues in academic libraries is essential. This includes topics such as copyright infringement, licenses, fair use practices, data privacy and security, and AI policies in academic libraries.

Conclusion

The study has successfully highlighted the current practices in academic libraries regarding AI adoption. Studies have also revealed the technical, institutional, environmental, and individual challenges that academic libraries continue to face due to AI adoption and their consequent use for information services in academic libraries globally. These findings are expected to foster relevant research that would culminate in the full application of AI in academic libraries and the provision of AI-related services in libraries. Having revealed the current practices, this systematic literature review identifies a cluster of challenges hindering AI adoption in academic libraries and proposes various future research directions for researchers. It contributes to knowledge and bridges a gap that previously did not exist in the literature. Given the importance of AI, this review's findings have significant implications for academic librarians, universities, academic libraries, and researchers who mostly rely on accurate and adequate access to information to thrive and meet their respective goals. With future research directions emphasizing the need for studies on full-scale adoption of AI tools in libraries (Ali et al., 2021), academic librarians and other information professionals must prioritize AI-related research, especially as academic libraries have shown, based on the review, great promise toward full adoption and provision of AI-related services in the foreseeable future.

Limitations of the study

This study has methodological and language limitations. While the adoption of a conceptual method was necessary to systematically identify current practices and present relevant articles that would be useful for future AI research in information science and academic libraries, a mixed-method approach that includes primary data would have offered more insights. In terms of language, only articles written in English were included. This constitutes a limitation, given that AI is ubiquitous; hence, some articles on current practices regarding AI adoption and the challenges in academic libraries may have been written in other languages. Therefore, related systematic reviews on studies written in languages other than English would remarkably contribute to the literature. The focus on studies published in the last five years (2020–2024) was intentional because of the proliferation of AI-related research in recent years. Nevertheless, more extensive reviews would address this limitation and foster continued discussion about the adoption of AI in academic libraries globally.

Practical implications

Based on the findings, several practical implications emerge that are crucial for advancing academic library practices in the era of AI. The integration of AI technologies into academic libraries has revamped the way library services are conceptualized, delivered, and managed. For librarians and information professionals, the findings signal a need to reimagine traditional technical and user services. Specifically, the study reveals that cataloguing, classification, and indexing are increasingly being enhanced through AI tools including Gen AI. The findings imply that AI has significantly helped to reduce human error and workload, which enables librarians to redirect their efforts toward more complex, user-centered tasks. The systematic literature review further indicates the growing application of AI in reference and information services, such as chat-based interactions, ready-reference queries, and personalized recommendations. This reflects the importance of retraining librarians to develop new competencies in managing AI-enhanced communication

tools, including chatbot integration and prompt engineering. The use of AI in enhancing information discovery, search optimization, and personalized learning resources also implies that library staff must become adept at using AI to support research consultation and user instruction. Academic libraries are therefore encouraged to design structured AI literacy programs both for staff and users. This can be done by incorporating ethical use, critical evaluation of AI-generated content, and creating awareness of AI limitations like hallucinations or misinformation.

In terms of administrative and circulation services, the findings suggest that AI tools are facilitating the automation of tasks such as self-check-out, data analytics, security management, and decision-making support. These developments imply that academic libraries need to realign their operational strategies to accommodate AI in managing user flow, resource tracking, and service efficiency. The findings further imply that incorporating facial recognition, RFID technology, and intelligent dashboards for workload management demands a shift toward technology-driven infrastructure and service planning. Moreover, since funding and infrastructural challenges remain major institutional barriers, library administrators must advocate for strategic investments in AI infrastructure and training through policy alignment and institutional support.

From a user education perspective, AI is playing a significant role in promoting information literacy, supporting orientation programs, and hosting AI-awareness initiatives. This reinforces the need for librarians to assume a proactive role as facilitators of ethical AI adoption by guiding students and faculty on responsible AI use in academic contexts. Additionally, the findings reveal that professional development and collaboration are enhanced when librarians engage with global AI research trends and digital knowledge-sharing platforms. As such, there is a strong call for continuous capacity building, peer collaboration, and participation in AI-centric networks to remain current and competitive in service delivery.

CRediT authorship contribution statement

Lateef Ayinde: Writing – review & editing, Writing – original draft, Methodology, Conceptualization. **Raphael Ebiefung:** Writing – review & editing, Writing – original draft, Validation. **Bolaji David Oladokun:** Writing – review & editing, Writing – original draft, Validation.

Declaration of competing interest

The authors contributed to all aspects of the research and take responsibility for its content.

Data availability

Data will be made available on request.

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